Exercise 4 : Union & Union ALL

1.
```
SELECT account_id, account_holder, 'Sandton' AS City
FROM branch_sandton_accounts
UNION
SELECT account_id, account_holder,
'Rosebank' AS city
FROM branch_rosebank_accounts;
```

Account_id	Account_holder	City
A001	Nomvula Dlamini	Sandton
A002	David Mokoena	Sandton
A003	Lerato Sithole	Sandton
A004	Peter Nkosi	Sandton
A005	Zanele Khumalo	Rosebank
A006	Thabo Motha	Rosebank

2.
```
SELECT product_code, product_name,
~~'Product_type'~~ 'Savings' AS product_type
FROM savings_products
UNION
SELECT product_code, product_name,
'Current' AS product_type
FROM current_products
```

Product_code	Product_name	Product_type
SV01	Basic Savings	Savings
SV02	Premium Savings	Savings
SV03	Youth Savings	Savings
SV04	Business Saving	Savings
CR01	Standard Current	Current
CR02	Gold Current	Current
CR03	Business Current	Current

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3.
```
SELECT staff_id, staff_name,
'retail' AS email
FROM retail_banking_staff
UNION
SELECT staff_id, staff_name,
'Corporate' AS email
FROM corporate_banking_staff;
```

staff_id	staff_name	email
S001	Mpho Radebe	NULL
S002	Brian Tshabalala	NULL
S003	Asha Patel	NULL
S004	Kabelo Mokoena	NULL
S005	Nandi Dube	NULL
S006	Sipho Khumalo	NULL

4.
```
SELECT city_code, city_name,
'Mobile' AS region
FROM mobile_branch_cities
SELECT city_code, city_name,
'digital', AS region
FROM digital_branch_cities
```

city_code	city_name	region
C01	Johannesburg	Mobile
C02	Pretoria	Mobile
C03	Cape Town	Mobile
C04	Durban	Mobile
C05	Polokwane	Digital
C06	Port Elizabeth	Mobile

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5. SELECT customer_id, customer_name,
'Push Notification' AS segment
FROM push_notification_targets
UNION
SELECT customer_id, customer_name,
'In-App Banner' AS segment
FROM inapp_banner_targets;

Customer_id	Customer_name	Segment
C1001	Nomsa Zwane	Push Notification
C1002	Andile Buthelezi	Push Notification
C1003	Fatima Mahomed	Push Notification
C1004	Ryno van Zyl	Push Notification
C1005	Thandeka Cele	In-App Banner
C1006	Samuel Nkosi	In-App Banner

6. SELECT transaction_id, amount_id, amount,
'atm01' AS transaction_date
FROM atm01_transactions
UNION ALL
SELECT transaction_id, amount_id, amount,
'atm02' AS transaction_date
FROM atm02_transactions;

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Transaction_id	Account_id	Amount	Transaction_date
T1001	A001	500.00	NULL
T1002	A002	1200.00	NULL
T1003	A001	300.00	NULL
T1004	A003	750.00	NULL
T1003	A001	300.00	NULL
T1005	A004	900.00	NULL
T1006	A002	450.00	NULL
T1007	A005	150.00	NULL

2. SELECT app_id, customer_id, loan_type,
'Gauteng' AS amount_requested
FROM gauteng_loan_applications
UNION ALL
SELECT app_id, customer_id, loan_type,
'Western_Cape' AS amount_requested
FROM Western_Cape_loan_applications;

Application_id	Customer_id	loan_type	Amount_requested
LA001	C1001	Home loan	NULL
LA002	C1002	Vehicle loan	NULL
LA003	C1003	Personal	NULL
LA004	C1004	Home loan	NULL
LA003	C1003	Personal	NULL
LA005	C1005	Vehicle loan	NULL
LA006	C1006	Business	NULL
LA007	C1007	Home loan	NULL

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8. SELECT complaint_id, customer_id, NULL AS category,
NULL AS logged_date
FROM email_complaints
UNION ALL
SELECT complaint_id, customer_id, NULL AS category,
NULL AS logged_date
FROM app_complaints;

Complaint_id	Customer_id	Category	logged_date
EC001	C2001	NULL	NULL
EC002	C2002	NULL	NULL
EC003	C2003	NULL	NULL
EC004	C2004	NULL	NULL
AC001	C2005	NULL	NULL
AC002	C2001	NULL	NULL
AC003	C2006	NULL	NULL
AC004	C2007	NULL	NULL

9. SELECT payment_id, account_id, amount,
NULL AS payment_date
FROM april_payments
UNION ALL
SELECT payment_id, account_id, amount,
NULL AS payment_date
FROM may_payments

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Payment-id	Account-id	Amount	Paymentdate
PAY001	A001	12500.00	NULL
PAY002	A002	4200.00	NULL
PAY003	A003	9200.00	NULL
PAY004	A004	3300.00	NULL
PAY005	A001	12500.00	NULL
PAY006	A005	7600.00	NULL
PAY007	A002	5100.00	NULL
PAY008	A006	2800.00	NULL

10. SELECT entry_id, account_id, entry_type, NULL as amount,
NULL as entry_date
FROM debit_entries
UNION ALL
SELECT entry_id, account_id, entry_type, NULL AS amount,
NULL AS entry_date
FROM credit_entries.

entry_id	account_id	entry_type	amount	entrydate
DR001	A001	Debit	NULL	NULL
DR002	A002	Debit	NULL	NULL
DR003	A003	Debit	NULL	NULL
DR004	A004	Debit	NULL	NULL
CR001	A001	Credit	NULL	NULL
CR002	A005	credit	NULL	NULL
CR003	A002	Credit	NULL	NULL
CR004	A006	Credit	NULL	NULL

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Bonus questions

1. Union, because it returns unique values and removes duplicates.
2. Union ALL, because duplicates are taken into consideration.
3. The query fails because UNION requires both SELECT statements to return the same number of columns, in the same order, with compatible data types.

fix: SELECT account_id, account_holder, 'Sandton' AS city,
FROM branch_sandton_accounts
UNION
SELECT account_id, account_holder, 'Rosebank' AS city
FROM branch_rosebank_accounts;

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